# BECS 22623
# Implementation of Numerical Methods using Appropriate software

**Department of Mathematics**
**Faculty of Science**
**University of Kelaniya**

# Introduction

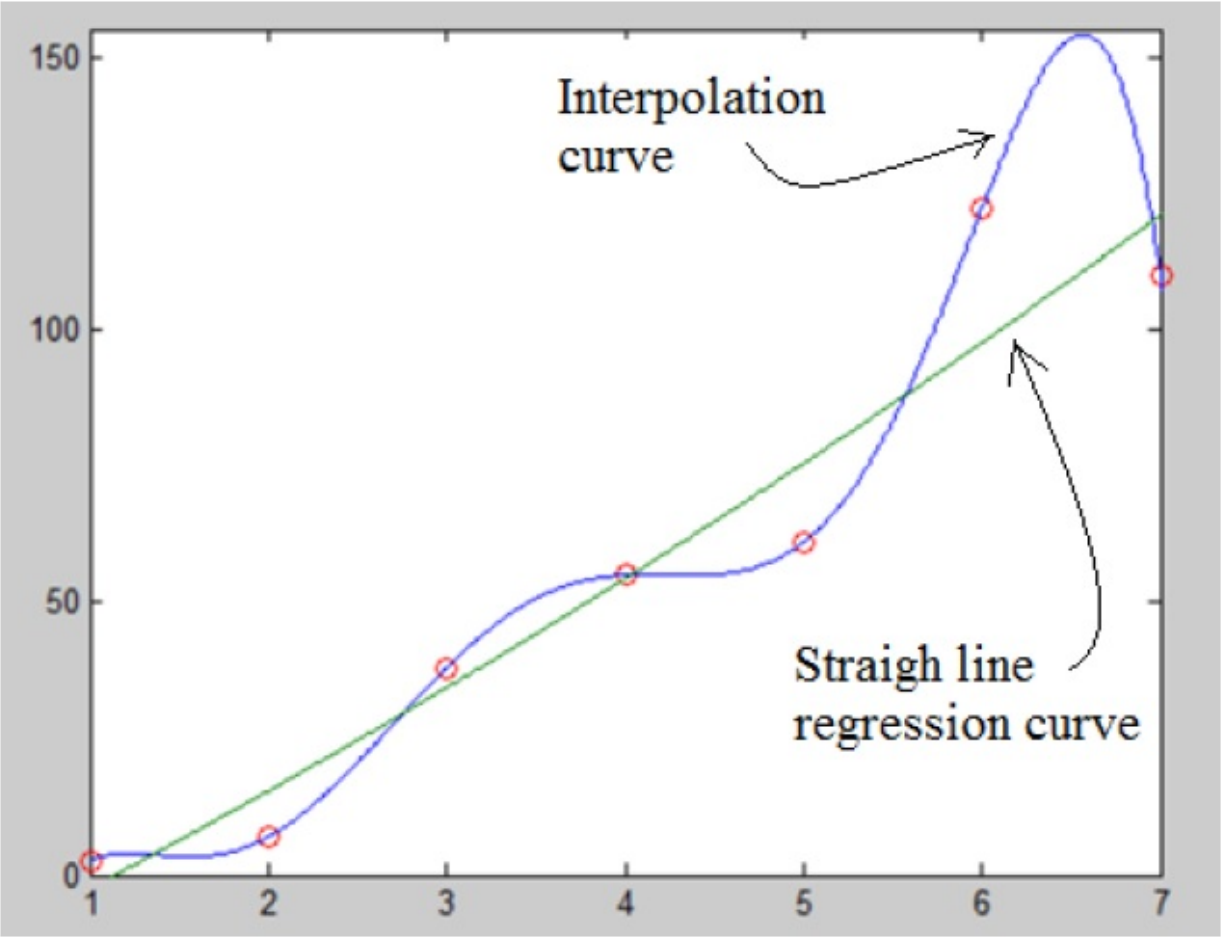
In the previous lecture, polynomial and cubic spline interpolation methods were introduced for estimating a value between a given set of precise data points.

The idea was to (interpolate) "fit" a function to the data points so as to perfectly pass through all data points.

In many real world problems the objective would be to find the best-fit analytic curve (model) that approximates the underlying functional relationship present in the data set.

Here, the best-fit curve is not required to pass through the data points, but it is required to capture the shape (general trend) of the data. This curve fitting problem is referred to as regression.

# Introduction



# LINEAR LEAST-SQUARES REGRESSION

The regression problem will be illustrated for fitting the linear model (straight line),$y(x) = a_1 x + a_0$ , to a set of $n$ paired experimental observations: $(x_1, y_1), (x_2, y_2), ..., (x_n, y_n)$.

**Idea:**

position the straight-line (i.e., to determine the regression coefficients $a_0$ and $a_1$) so that some *error measure* of fit is minimized.

A common error measure is the sum-of-the-squares (SSE) of the residual errors $e_i = y_i - y(x_i)$,

$$E(a_0, a_1) = \sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} [y_i - (a_1 x_i + a_0)]^2$$

# LINEAR LEAST-SQUARES REGRESSION

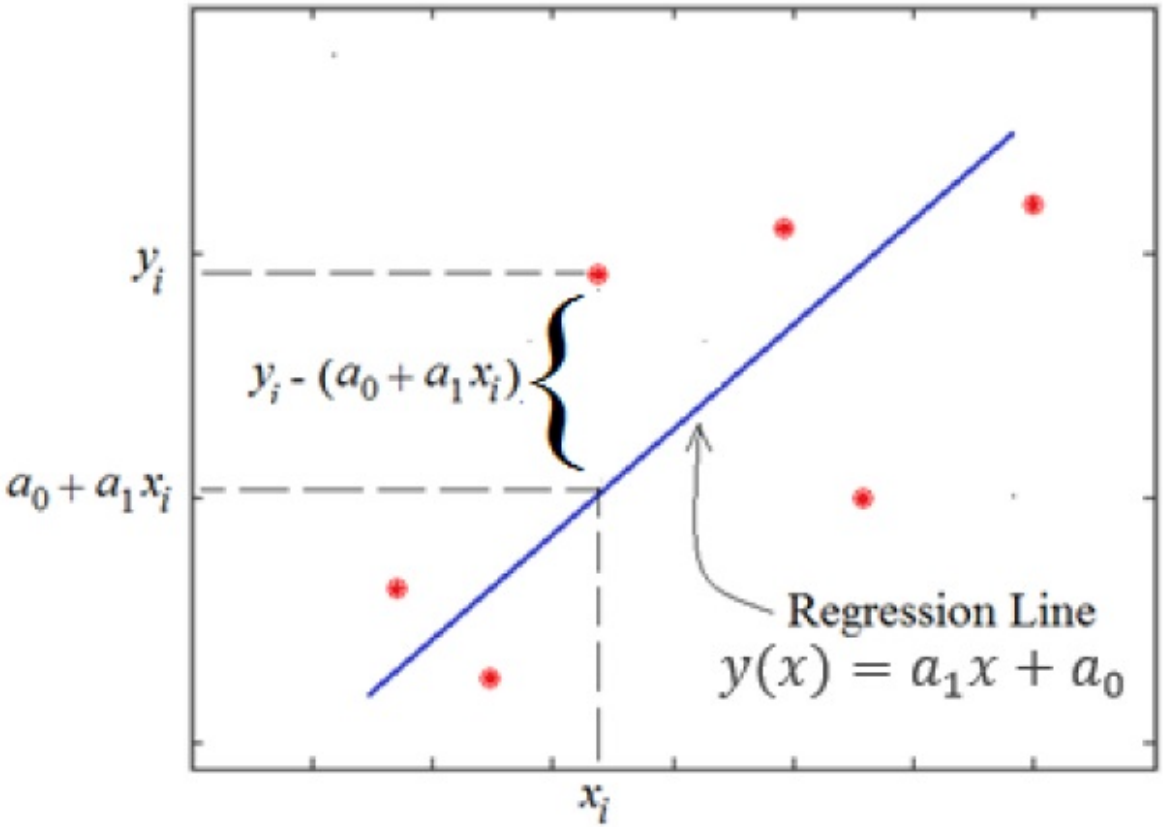



Figure: The residual error for the $i$th data point.

$E$ is known as *least-squares-error (LSE) criterion*

# LINEAR LEAST-SQUARES REGRESSION

A solution can be obtained for the regression coefficients, $\{a_1, a_0\}$, that minimizes $E(a_0, a_1)$.

To determine values for $a_0$ and $a_1$,
$E(a_0, a_1) = \sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} [y_i - (a_1 x_i + a_0)]^2$ is differentiated with respect to each unknown coefficient:

Setting these derivatives equal to zero will result in a minimum $E(a_0, a_1)$.

Some computations are included however for you to better understand the concepts.

# LINEAR LEAST-SQUARES REGRESSION

$$\frac{\partial}{\partial a_0} E(a_0, a_1) = 2 \sum_{i=1}^{n} (y_i - a_1 x_i - a_0)(-1) = 0$$

$$\frac{\partial}{\partial a_1} E(a_0, a_1) = 2 \sum_{i=1}^{n} (y_i - a_1 x_i - a_0)(-x_i) = 0$$

$$-\sum_{i=1}^{n} y_i + \sum_{i=1}^{n} a_0 + \sum_{i=1}^{n} a_1 x_i = 0$$

$$-\sum_{i=1}^{n} x_i y_i + \sum_{i=1}^{n} a_0 x_i + \sum_{i=1}^{n} a_1 x_i^2 = 0$$

# LINEAR LEAST-SQUARES REGRESSION

Now, realizing that $\sum a_0 = na_0$, we can express the equations as a set of two simultaneous linear equations with two unknowns ($a_0$ and $a_1$):

$$n \quad a_0 + \left(\sum x_i\right) a_1 = \sum y_i$$

$$\left(\sum x_i\right) a_0 + \left(\sum x_i^2\right) a_1 = \sum x_i y_i$$

These are called the *normal equations*. They can be solved simultaneously for

$$a_1 = \frac{n \sum x_i y_i - \sum x_i \sum y_i}{n \sum x_i^2 - \left(\sum x_i\right)^2}$$

# LINEAR LEAST-SQUARES REGRESSION

## Example

Fit a straight line to the values in the following table: which contains experimental data for force (N) and velocity (m/s) from a wind tunnel experiment.

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| $v$, m/s | 10 | 20 | 30 | 40 | 50 | 60 | 70 | 80 |
| $F$, N | 25 | 70 | 380 | 550 | 610 | 1220 | 830 | 1450 |

# LINEAR LEAST-SQUARES REGRESSION

In this application, force is the dependent variable ($y$) and velocity is the independent variable ($x$). The data can be set up in tabular form and the necessary sums computed as in the following table:

| $i$ | $x_i$ | $y_i$ | $x_i^2$ | $x_i y_i$ |
|---|---|---|---|---|
| 1 | 10 | 25 | 100 | 250 |
| 2 | 20 | 70 | 400 | 1,400 |
| 3 | 30 | 380 | 900 | 11,400 |
| 4 | 40 | 550 | 1,600 | 22,000 |
| 5 | 50 | 610 | 2,500 | 30,500 |
| 6 | 60 | 1,220 | 3,600 | 73,200 |
| 7 | 70 | 830 | 4,900 | 58,100 |
| 8 | 80 | 1,450 | 6,400 | 116,000 |
| $\sum$ | 360 | 5,135 | 20,400 | 312,850 |

# LINEAR LEAST-SQUARES REGRESSION

The means can be computed as

$$\overline{x} = \frac{360}{8} = 45, \qquad \overline{y} = \frac{5135}{8} = 641.875$$

The slope and the intercept can then be calculated,

$$a_1 = \frac{8(312{,}850) - 360(5{,}135)}{8(20{,}400) - (360)^2} = 19.47024$$

$$a_0 = 641.875 - 19.47024(45) = -234.2857$$

Using force and velocity in place of $y$ and $x$, the least-squares fit is

$$F = -234.2857 + 19.47024v$$

# LINEAR LEAST-SQUARES REGRESSION

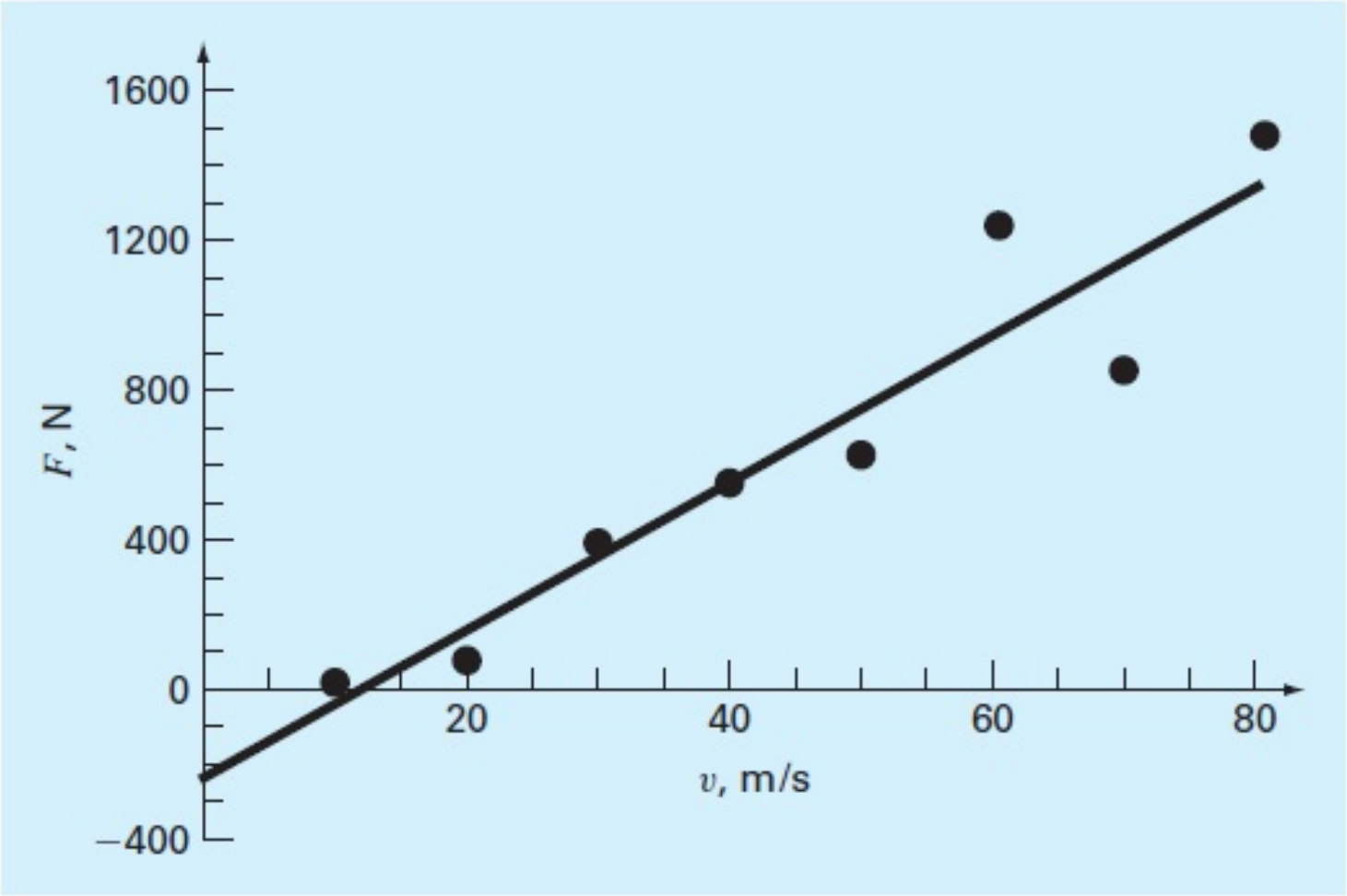



Figure: Least-squares fit of a straight line to the data from our example.

# LINEAR LEAST-SQUARES REGRESSION | Quantification of Error of Linear Regression

Recall that

$$E(a_0, a_1) = \sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} [y_i - (a_1 x_i + a_0)]^2$$

Let $E_M$ be the total sum of the squares of the residuals between the data points and the mean,

$$E_M = \sum_{i=1}^{n} (y_i - \overline{y})^2$$

Then, the (positive) difference $E_M - E$ represents the improvement (where the smaller $E$ is, the better) due to describing the data in terms of a straight-line, rather than as an average value (a straight-line with zero slope and $y$-intercept equals to $\overline{y}$.

# LINEAR LEAST-SQUARES REGRESSION | Quantification of Error of Linear Regression

The coefficient of determination, $r^2$ is defined as the relative error between $E_M$ and $E$:

$$r^2 = \frac{E_m - E}{E_M} = 1 - \frac{E}{E_M}$$

For perfect fit, where the regression line goes through all data points, $E = 0$ and $r^2 = 1$.

On the other hand for $E_M = E$, we have $r^2 = 0$, and the fit represents no improvement over a simple average. A value of $r^2$ between 0 and 1

represents the extent of improvement.

$r$- correlation coefficient

# LINEAR LEAST-SQUARES REGRESSION

## Example

Fit a straight-line to the data provided in the following table. Find $r^2$.

| $x$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|---|
| $y$ | 2.5 | 7 | 38 | 55 | 61 | 122 | 110 |

# LINEAR LEAST-SQUARES REGRESSION

**Solution:** The following Matlab script computes the linear regression coefficients:

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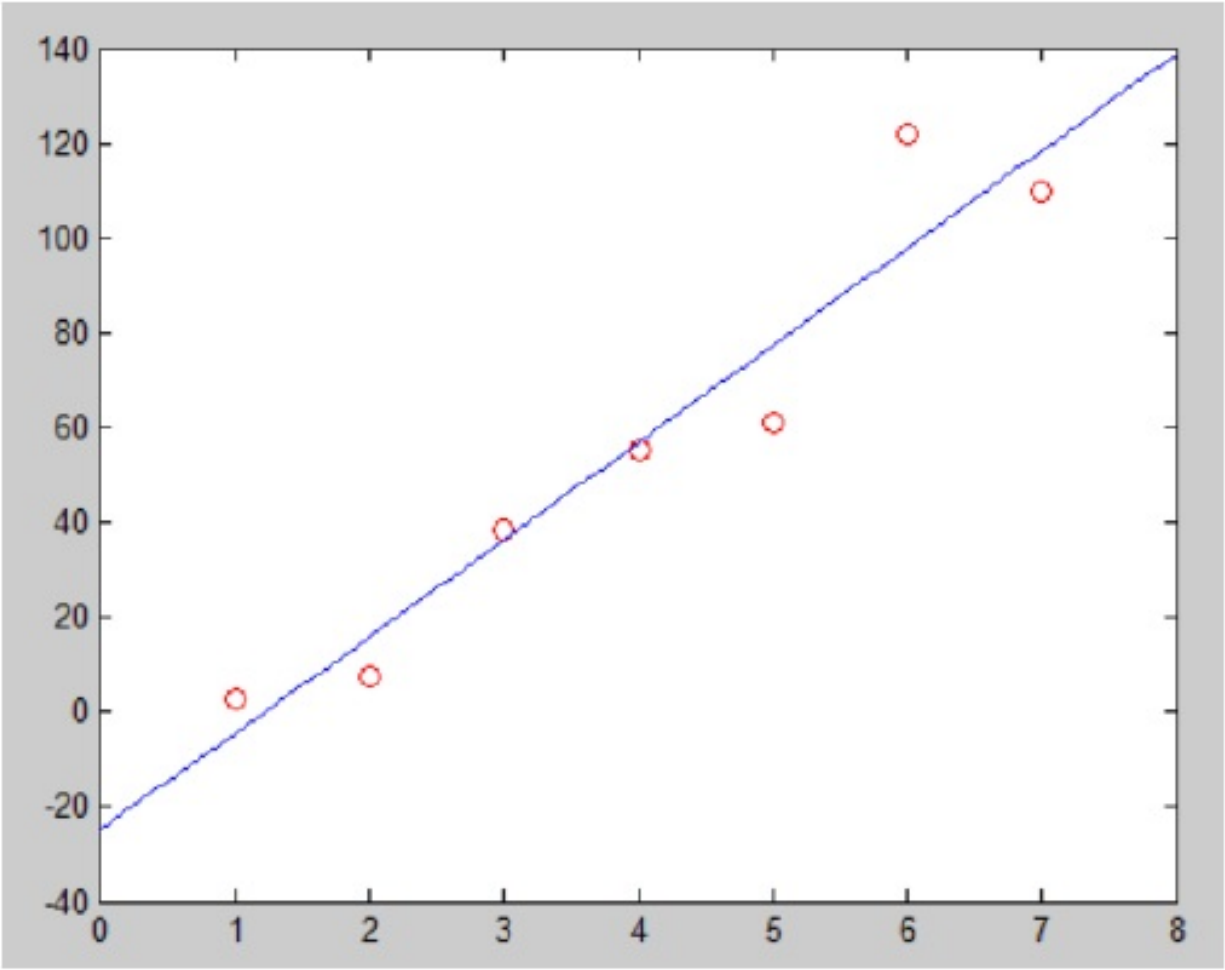
```
>> x=[1 2 3 4 5 6 7];
y=[2.5 7 38 55 61 122 110];
n=length(x);
a1=(n*sum(x.*y)-sum(x)*sum(y))/(n*sum(x.^2)-(sum(x)).^2)
a0=sum(y)/n-a1*sum(x)/n

a1 =

   20.5536


a0 =

  -25.7143

>>
```

# LINEAR LEAST-SQUARES REGRESSION

**Solution:** The following plot displays the data and the regression model, $y(x) = 20.5536x - 25.7143$.



# LINEAR LEAST-SQUARES REGRESSION

**Solution:** The following script computes the correlation coefficient, $r$.

```
>> x=[1 2 3 4 5 6 7];
y=[2.5 7 38 55 61 122 110];
n=length(x);
r=(n*sum(x.*y)-sum(x)*sum(y))/((sqrt(n*sum(x.^2)-(sum(x))^2))*(sqrt(n*sum(y.^2)-(sum(y))^2)))

r =

    0.9582
```

So, $r^2 = 0.9181$. These results indicate that about 92% of the variability in the data has been explained by the linear model.

# LINEAR LEAST-SQUARES REGRESSION | MATLAB
## M-file: **linregr**

An algorithm for linear regression can be easily developed. The required summations are readily computed with MATLAB's **sum** function.

# LINEAR LEAST-SQUARES REGRESSION | MATLAB

## M-file: **linregr**

```
function [a, r2] = linregr(x,y)
% linregr: linear regression curve fitting
%    [a, r2] = linregr(x,y):Least squares fit of straight
%            line to data by solving the normal equations

% input:
%   x = independent variable
%   y = dependent variable
% output:
%   a = vector of slope, a(1), and intercept, a(2)
%   r2 = coefficient of determination

n = length(x);
if length(y)~=n, error('x and y must be same length'); end
x = x(:); y = y(:);     % convert to column vectors
sx = sum(x); sy = sum(y);
sx2 = sum(x.*x); sxy = sum(x.*y); sy2 = sum(y.*y);
a(1) = (n*sxy-sx*sy)/(n*sx2-sx^2);
a(2) = sy/n-a(1)*sx/n;
r2 = ((n*sxy-sx*sy)/sqrt(n*sx2-sx^2)/sqrt(n*sy2-sy^2))^2;
% create plot of data and best fit line
xp = linspace(min(x),max(x),2);
yp = a(1)*xp+a(2);
plot(x,y,'o',xp,yp)
```

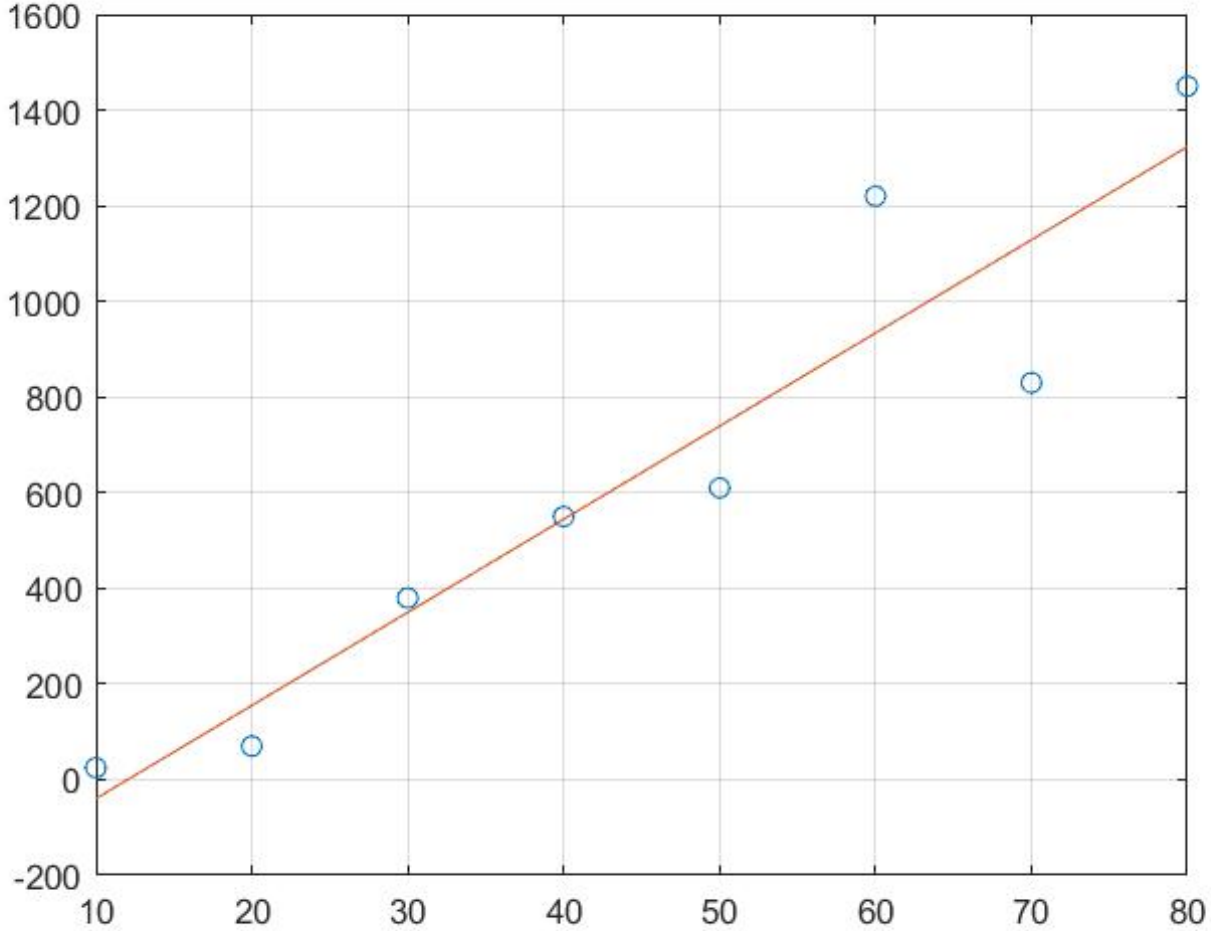
# LINEAR LEAST-SQUARES REGRESSION | MATLAB
## M-file: **linregr**

A simple example of the use of this M-file would be to fit the force-velocity data analyzed in our first example:

```
>> x = [10 20 30 40 50 60 70 80];
>> y = [25 70 380 550 610 1220 830 1450];
>> linregr(x,y)

r2 =
    0.8805

ans =
   19.4702 -234.2857
```

# LINEAR LEAST-SQUARES REGRESSION | MATLAB

## M-file: **linregr**



# LINEAR LEAST-SQUARES REGRESSION

## Problem

Use least-squares regression to fit a straight line to

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| $x$ | 0 | 2 | 4 | 6 | 9 | 11 | 12 | 15 | 17 | 19 |
| $y$ | 5 | 6 | 7 | 6 | 9 | 8 | 8 | 10 | 12 | 12 |

Along with the slope and intercept, compute the standard error of the estimate and the correlation coefficient. Plot the data and the regression line. Then repeat the problem, but regress $x$ versus $y$, that is, switch the variables. Interpret your results.

# LINEAR LEAST-SQUARES REGRESSION

## Problem

You perform experiments and determine the following values of heat capacity $c$ at various temperatures $T$ for a gas:

| $T$ | −50 | −30 | 0 | 60 | 90 | 110 |
|---|---|---|---|---|---|---|
| $c$ | 1250 | 1280 | 1350 | 1480 | 1580 | 1700 |

Use regression to determine a model to predict $c$ as a function of $T$.

# Linearization of Nonlinear Models

The straight-line regression model is not always suitable for curve fitting. The choice of regression model is often guided by the plot of the available data, or can be guided by the knowledge of the physical behavior of the system that generated the data.

In general, polynomial or other nonlinear models are more suitable.

However, some basic nonlinear functions can be readily transformed into linear functions in their regression coefficients (we will refer to such functions as **transformable** or **linearizable**).

# Linearization of Nonlinear Models | Linearizable linear model

The *exponential model*:

$$y = \alpha_1 e^{\beta_1 x}$$

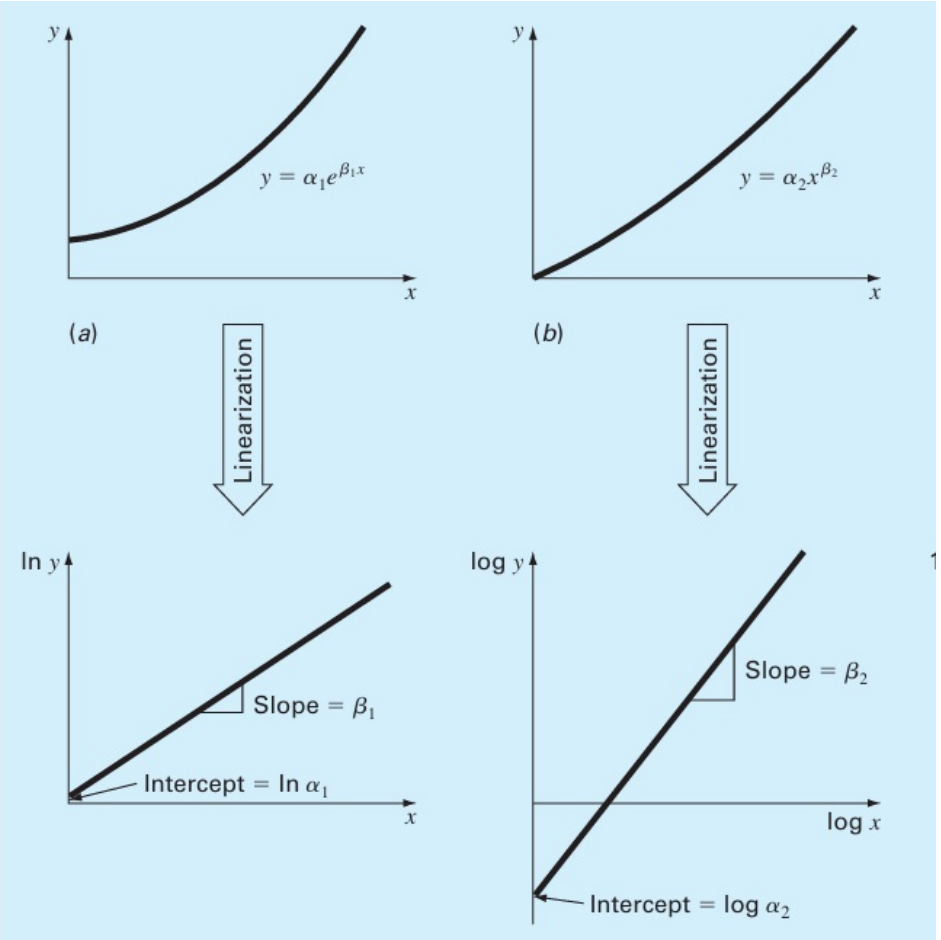
where $\alpha_1$ and $\beta_1$ are constants.

The *exponential model* can be linearized by taking its natural logarithm to yield

$$\ln y = \ln \alpha_1 + \beta_1 x$$

Thus, a plot of $\ln y$ versus $x$ will yield a straight line with a slope of $\beta_1$ and an intercept of $\ln \alpha_1$.

# Linearization of Nonlinear Models | Linearizable linear model



# Linearization of Nonlinear Models | Linearizable linear model

## Example

Fit the exponential model to the data in the following table. Compare the fit quality to that of the straight-line model.

| $x$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| $y$ | 2.5 | 7 | 38 | 55 | 61 | 122 | 83 | 143 |

# Linearization of Nonlinear Models | Linearizable linear model

Using our M-file **linregr** we see that the linear model solution is $y = 19.3036x - 22.9286$, $r^2 = 0.8811$.
For the exponential model:

```
>> x=[1 2 3 4 5 6 7 8];
y=[2.5 7 38 55 61 122 83 143];
ye=log(y);
n=length(x);
a1=(n*sum(x.*ye)-sum(x)*sum(ye))/(n*sum(x.^2)-(sum(x)).^2);
a0=sum(ye)/n-a1*sum(x)/n;
r=(n*sum(x.*ye)-sum(x)*sum(ye))/((sqrt(n*sum(x.^2)-(sum(x))^2))*(sqrt(n*sum(ye.^2)-(sum(ye))^2)));
alpha=exp(a0), beta=a1, r^2

alpha =

    3.4130


beta =

    0.5273


ans =

    0.8141
```

# MATLAB Functions: polyfit

MATLAB has a built-in function **polyfit** that fits a least-squares nth-order polynomial to data. It can be applied as in:

$$>> \text{p} = \text{polyfit}(x, y, n)$$

where $x$ and $y$ are the vectors of the independent and the dependent variables, respectively, and $n$ is the order of the polynomial.

The function returns a vector $p$ containing the polynomial's coefficients. We should note that it represents the polynomial using decreasing powers of $x$.

**Remark:** polyfit$(x, y, 1)$ will return the slope and the intercept of the best-fit straight line.

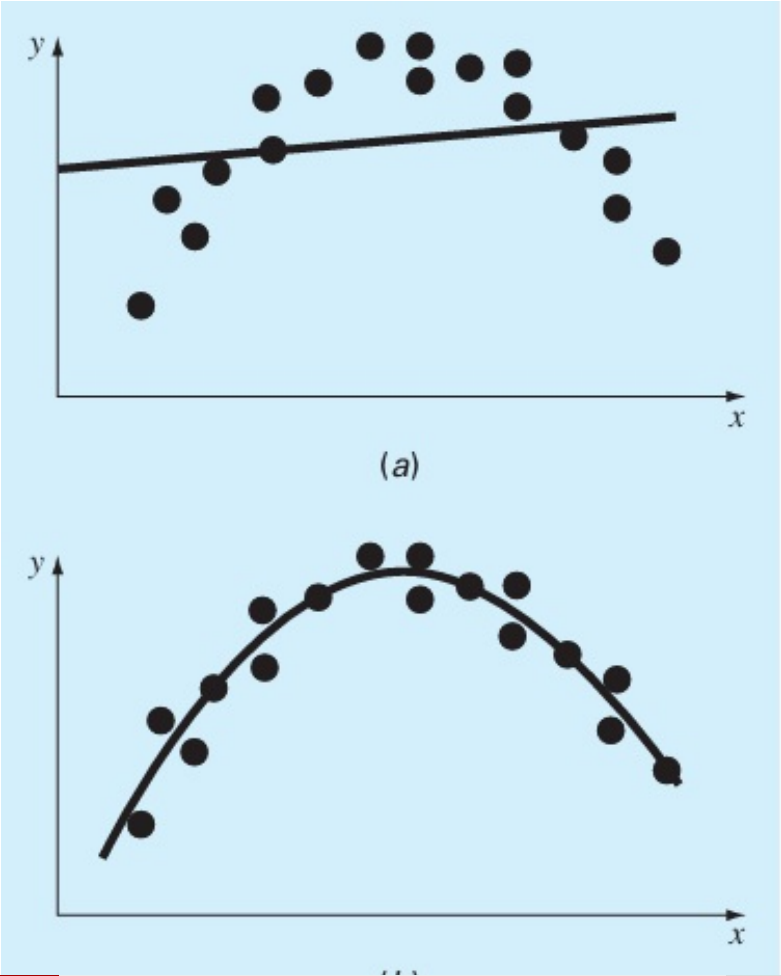
# MATLAB Functions: polyfit

```
>> x = [10 20 30 40 50 60 70 80];
>> y = [25 70 380 550 610 1220 830 1450];
>> a = polyfit(x,y,1)

a =
   19.4702 -234.2857
```

Thus, the slope is 19.4702 and the intercept is $-234.2857$.

# POLYNOMIAL REGRESSION

Some data, although exhibiting a marked pattern such as seen in the following figure, are poorly represented by a straight line. For these cases, a curve would be better suited to fit the data.



# POLYNOMIAL REGRESSION

There are two methods to handle this method.

- Transformation
- Polynomial Regression

# POLYNOMIAL REGRESSION

The sum of the squares of the residuals is

$$S_r = \sum_{i=1}^{n} \left(y_i - a_0 - a_1 x_i - a_2 x_i^2\right)^2$$

To generate the least-squares fit, we take the derivative of the above equation with respect to each of the unknown coefficients of the polynomial, as in

$$\begin{aligned}
(n)a_0 + \left(\sum x_i\right) a_1 + \left(\sum x_i^2\right) a_2 &= \sum y_i \\
\left(\sum x_i\right) a_0 + \left(\sum x_i^2\right) a_1 + \left(\sum x_i^3\right) a_2 &= \sum x_i y_i \\
\left(\sum x_i^2\right) a_0 + \left(\sum x_i^3\right) a_1 + \left(\sum x_i^4\right) a_2 &= \sum x_i^2 y_i
\end{aligned}$$

# POLYNOMIAL REGRESSION

These equations can be set equal to zero and rearranged to develop the following set of normal equations:

$$
\begin{aligned}
(n)a_0 + \left(\sum x_i\right) a_1 + \left(\sum x_i^2\right) a_2 &= \sum y_i \\
\left(\sum x_i\right) a_0 + \left(\sum x_i^2\right) a_1 + \left(\sum x_i^3\right) a_2 &= \sum x_i y_i \\
\left(\sum x_i^2\right) a_0 + \left(\sum x_i^3\right) a_1 + \left(\sum x_i^4\right) a_2 &= \sum x_i^2 y_i
\end{aligned}
$$

# POLYNOMIAL REGRESSION

**TABLE 15.1** Computations for an error analysis of the quadratic least-squares fit.

| $x_i$ | $y_i$ | $(y_i - \bar{y})^2$ | $(y_i - a_0 - a_1 x_i - a_2 x_i^2)^2$ |
|---|---|---|---|
| 0 | 2.1 | 544.44 | 0.14332 |
| 1 | 7.7 | 314.47 | 1.00286 |
| 2 | 13.6 | 140.03 | 1.08160 |
| 3 | 27.2 | 3.12 | 0.80487 |
| 4 | 40.9 | 239.22 | 0.61959 |
| 5 | 61.1 | 1272.11 | 0.09434 |
| $\sum$ | 152.6 | 2513.39 | 3.74657 |

Solution. The following can be computed from the data:

$$m = 2 \qquad \sum x_i = 15 \qquad \sum x_i^4 = 979$$

$$n = 6 \qquad \sum y_i = 152.6 \qquad \sum x_i y_i = 585.6$$

$$\bar{x} = 2.5 \qquad \sum x_i^2 = 55 \qquad \sum x_i^2 y_i = 2488.8$$

$$\bar{y} = 25.433 \qquad \sum x_i^3 = 225$$

# POLYNOMIAL REGRESSION

Therefore, the simultaneous linear equations are

$$\begin{bmatrix} 6 & 15 & 55 \\ 15 & 55 & 225 \\ 55 & 225 & 979 \end{bmatrix} \begin{Bmatrix} a_0 \\ a_1 \\ a_2 \end{Bmatrix} = \begin{Bmatrix} 152.6 \\ 585.6 \\ 2488.8 \end{Bmatrix}$$

These equations can be solved to evaluate the coefficients. For example, using MATLAB:

```
>> N = [6 15 55;15 55 225;55 225 979];
>> r = [152.6 585.6 2488.8];
>> a = N\r

a =
    2.4786
    2.3593
    1.8607
```

Therefore, the least-squares quadratic equation for this case is

$$y = 2.4786 + 2.3593x + 1.8607x^2$$

The standard error of the estimate based on the regression polynomial is [Eq. (15.3)]

$$s_{y/x} = \sqrt{\frac{3.74657}{6-(2+1)}} = 1.1175$$

The coefficient of determination is

$$r^2 = \frac{2513.39 - 3.74657}{2513.39} = 0.99851$$

and the correlation coefficient is $r = 0.99925$.

# NONLINEAR REGRESSION

There are many cases in engineering and science where nonlinear models must be fit to data. In the present context, these models are defined as those that have a nonlinear dependence on their parameters.

For an example $y = a_0(1 - e^{-a_1 x})$.

As with linear least squares, nonlinear regression is based on determining the values of the parameters that minimize the sum of the squares of the residuals. However, for the nonlinear case, the solution must proceed in an iterative fashion.

An alternative is to use optimization techniques to directly determine the least-squares fit. For the above example, it can be expressed as an objective function to compute the sum of the squares:

$$f(a_0, a_1) = \sum_{i=1}^{n} [y_i - a_0(1 - e^{-a_1 x_i})]^2$$

# MATLAB's fminsearch function

**Matlab's fminsearch:**

Matlab has an important built-in function for numerical minimization of nonlinear multivariable functions. The function name is **fminsearch**.

Syntax: fminsearch(fun,x0)

fun- function that you need to evaluate, x0- a vector of initial values.

# NONLINEAR REGRESSION

Consider

$$y(x) = a_0(1 - e^{-a_1 x})$$

```
function f=fss(a)

  x=[-2 0 2 4];
  y=[1 0 -4 -12];
  yp=a(1)*(1-exp(-(a(2)*x)));
  f=sum((y-yp).^2);
end
```

```
Command Window
>> fminsearch(@fss,[0 0])

ans =

  2.497930060098661  -0.441002312329933
```

We have, the solution $y(x) = 2.498(1 - e^{-0.441x})$.

# NONLINEAR REGRESSION

## Problem

It is known that the data tabulated below can be modeled by the following equation

$$y = \left(\frac{a + \sqrt{x}}{b\sqrt{x}}\right)^2$$

| | | | | | |
|---|---|---|---|---|---|
| $x$ | 0.5 | 1 | 2 | 3 | 4 |
| $y$ | 10.4 | 5.8 | 3.3 | 2.4 | 2 |

Use nonlinear regression to determine the parameters $a$ and $b$. Based on your analysis predict $y$ at $x = 1.6$.

# NONLINEAR REGRESSION

```
function f=nonlireg(a)
  x=[0.5 1 2 3 4];
  y=[10.4 5.8 3.3 2.4 2];
  yp=((a(1)*sqrt(x))/(a(2)*sqrt(x))).^2;
  f=sum((y-yp).^2);
end
```

Command Window

```
>> fminsearch(@nonlireg,[1 1])

ans =

  1.336842561466255   0.611457385682849
```